PRODUCT PROSPECTUS & MARKET ANALYSIS

Using Data to Drive Business Performance

Databases and Management Information Systems

Nim Dvir, PhD
Massry School of Business, UAlbany

Live Platform: https://database-book-project.vercel.app/

1. The Market Opportunity: Solving the "Two-Book Problem"

Currently, to teach a business-facing database course, instructors are forced to assign a **two-book bundle**, creating an inherent "stitching tax" on course design:

- **Book 1: Theory.** *Database Systems 14e (Coronel)* is used to teach relational theory, normalization, and database concepts, but proves too abstract and engineering-focused for non-CS students.
- **Book 2: Execution.** *Illustrated Microsoft 365 1e (Beskeen)* is used to teach students how to build forms and queries in Access, but functions as a software manual with no theoretical depth.

The Solution: *Using Data to Drive Business Performance* replaces this fragmented experience. It is a unified, single-textbook solution that integrates database theory directly with hands-on business execution, tailored specifically for the modern business undergraduate.

2. Head-to-Head Alignment Comparison

The Category Refresh: This manuscript is not positioned to cannibalize comprehensive engineering texts like *Coronel*. It is positioned as the modern, analytics-era successor to the applied slot currently occupied by **Concepts of Database Management, 10e** (2021). The goal is to modernize a category Cengage already leads, addressing modern cloud and SQL requirements before competitors capture the shifting curriculum.

Feature	Database Systems 14e *(Coronel)*	Illustrated M365 1e *(Beskeen)*	Using Data to Drive Business Performance
Primary Market Fit	CS and traditional IT/MIS majors.	Vocational, entry-level software training.	**Business Analytics, Finance, Marketing, and modern MIS majors.**
Pedagogical Sequence	**Theory-First:** Forces students to design complex ERDs from scratch before writing a query.	**Click-First:** Teaches where menus are, but ignores *why* databases are structured this way.	**The "Design Pivot":** Students learn to *query* first (as users) and *design* second (as creators).
SQL & Tech Depth	Deep, but highly theoretical. Treats SQL as mathematical set theory.	Almost non-existent. Relies on Access QBE with no modern SQL logic.	**Analytical SQL:** Teaches window functions, CTEs, and CASE logic for business questions.
Cloud & AI	Bolted on. Relegated to theoretical chapters at the end of the book.	None. Strictly localized, desktop-bound training.	**Native & Earned AI:** Integrates Supabase, cloud RLS, and vector/RAG architecture.
Business Strategy	Minimal. Focuses on system building, not business output.	None. Focuses on software competency.	**DIKW Pathway:** Closes the loop with BI reporting, KPI thresholds, and decision memos.

3. Key Differentiators & Pedagogical Innovation

The "Design Pivot"

Traditional texts front-load complex normalization (1NF-3NF) early on, demoralizing business students. This structure allows students to *use* data first—building confidence and context—before pivoting to *designing* schemas in Chapter 9.

Real SQL vs. QBE

While Access is a great starting point, modern employers demand real, executable SQL (the #1 requested data analyst skill). The text explicitly transitions students from local environments to modern dialects (PostgreSQL/SQLite).

"Earned AI" Architecture

Rather than slapping AI sidebars into old chapters, AI is taught as a structural consumer of databases—showing students that without clean schemas, AI hallucinations happen. It introduces modern vector/RAG concepts natively.

Major-Specific Career Translation

Chapter 16 explicitly maps database skills to specific business majors (Accounting, Finance, Marketing) and helps students translate lab work into resume-ready STAR interview answers—crucial for retention-focused business schools.

AI-Resistant Assessment

Built structurally post-ChatGPT. End-of-chapter problems aren't simple lookup questions; they are artifact-based (ERDs, normalized schemas, decision memos) applied to an unseen scenario (PetVax), solving the assignment integrity problem at the architectural level.

Revision Economics & Modularity

Tool-dependent content (Access, Supabase, BI) is quarantined to specific modules (Ch. 11, 12, 14, and appendices). The relational spine (Ch. 2–10) is tool-neutral, allowing for surgical, cost-effective updates rather than full revision cycles.

4. Curriculum Alignment & Market Demand

- **IS2020 Competency Crosswalk:** The ACM/AIS IS2020 guidelines mandate *Data and Information Management* as a required competency area. This manuscript directly fulfills the core mandates: Querying (Ch. 5/10), Design (Ch. 6/7/9), Security (Ch. 11/14), and Non-relational models (Ch. 14).

- **Combating OER Affordability Pressures:** Departments under budget pressure are increasingly abandoning expensive, disjointed bundles for free (but limited) OER texts. A cohesive, digital-first product provides the unified quality required to win those seats back into the publisher ecosystem.

- **A Tested, "Close-the-Loop" Architecture:** Built around a consistent six-file architecture (Core, Build, Review, Terms, RAT, Lab). Every concept is taught theoretically and then immediately executed practically in the context of the running *PetVax* and *Grading Database* projects.

The Strategic Imperative

"The ACM/AIS IS2020 curriculum guidelines mandate Data and Information Management as a required competency. Right now, business schools are struggling to fulfill that with heavy CS texts or lightweight software manuals. ***Using Data to Drive Business Performance*** *is built to capture the business students who need to understand databases as strategic business infrastructure, not just IT storage."*

Prepared Deliverables for Review:
- Complete 16-Chapter Outline
- Three Representative Chapters
- MS Access Appendix
- Live Prototype Access